

Question:

I have 10 liters of orange drink that are two-thirds water and I wish to add it to 15 liters of pineapple drink that is three-fifths water. But as I pour it, I spill one liter of the orange drink. How much water is in the remaining 24 liters?

Reasoning + Answer:

Let's break down the problem step by step to determine how much water is in the remaining 24 liters after the spill .

1 . **Calculate the amount of water in the original 10 liters of orange drink : **

- The orange drink is two-thirds water .
- To find the amount of water in 10 liters , we calculate :

$$\text{Amount of water in orange drink} = \frac{2}{3} \times 10$$

- Performing the multiplication :
$$\frac{2}{3} \times 10 = \frac{20}{3} \approx 6.67 \text{ liters}$$

2 . **Calculate the amount of water spilled : **

- One liter of the orange drink is spilled .
- To find the amount of water in this one liter , we use the fraction of water in the original mixture :

$$\text{Amount of water in spilled orange drink} = \frac{2}{3} \times 1$$

- Performing the multiplication :
$$\frac{2}{3} \times 1 = \frac{2}{3} \approx 0.67 \text{ liters}$$

3 . **Calculate the remaining amount of water in the orange drink after the spill : **

- Subtract the amount of water spilled from the original amount of water in the orange drink :

$$\text{Remaining water in orange drink} = \frac{20}{3} - \frac{2}{3}$$

- Perform the subtraction :
$$\frac{20}{3} - \frac{2}{3} = \frac{18}{3} = 6 \text{ liters}$$

4 . **Calculate the amount of water in the 15 liters of pineapple drink : **

- The pineapple drink is three-fifths water .
- To find the amount of water in 15 liters , we calculate :

$$\text{Amount of water in pineapple drink} = \frac{3}{5} \times 15$$

- Performing the multiplication :
$$\frac{3}{5} \times 15 = 9 \text{ liters}$$

5 . **Calculate the total amount of water in the remaining 24 liters : **

- Add the remaining water in the orange drink to the water in the pineapple drink :

$$\text{Total amount of water} = 6 + 9$$

- Perform the addition :
$$6 + 9 = 15 \text{ liters}$$

Thus , the total amount of water in the remaining 24 liters is 15 liters .

<think>
1 . Orange drink : $\frac{20}{3} - \frac{2}{3} = 6$ liters of water remaining
2 . Pine apple drink : $\frac{3}{5} * 15 = 9$ liters of water
3 . Total water : $6 + 9 = 15$ liters
</think>
<answer>15</answer>